Customer Segmentation using K-Means Clustering Detailed Project Report

Objective

Build customer segments using K-Means clustering to support targeted marketing.

Dataset

Marketing Campaign dataset containing customer demographics, income, purchases and campaign response.

Data Cleaning

Handled missing values, converted Dt_Customer to datetime, created Age, Children, Customer_Years and Total_Spending features.

EDA

Analyzed distributions, relationships and correlations using histograms, scatter plots, boxplots and a heatmap. Income generally increases with spending, while spending varies across customer groups.

Scaling

Applied StandardScaler to normalize numerical features before clustering.

PCA

Reduced dimensionality to two principal components for visualization while preserving major variance.

Elbow Method

Used inertia vs K plot to determine the optimal number of clusters.

K-Means

Clustered customers into meaningful groups based on purchasing behaviour.

Business Insights

High-value customers deserve loyalty programs; medium-value customers respond well to personalized recommendations; low-value customers can be targeted with discounts; family segments benefit from bundled offers.

Conclusion

The workflow demonstrates an end-to-end machine learning pipeline: cleaning, EDA, feature engineering, scaling, PCA, clustering and business interpretation.